## Ejercicio14_Hoja_1

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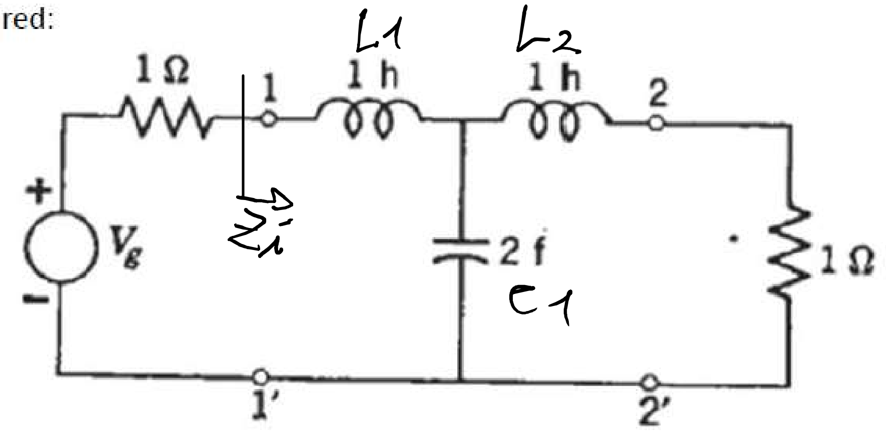
**Ejercicio 14.**

Para la siguiente red:




A) Calcular los parámetros S de la red.

B) Obtener analíticamente y graficar |S11(w)| y |S21(w)|. Extraer conclusiones acerca de los mismos en banda detenida y banda de atenuación

a) $Z_g = Z_L = Z_0 = R_g$

- $S_{11} = \dfrac{Z_i - Z_{01}}{Z_i + Z_{01}} \Rightarrow Z_i = \left[(sL_2 + Z_g) \,//\, \dfrac{1}{sC_1}\right] + \underbrace{sL_1}_{Z_{L1}}, \quad Z_{01} = Z_g$

$$Z_i = \left[\frac{1}{sL_2 + R_g} + sC_1\right]^{-1} + sL_1 = \left[\frac{1 + (sL_2 + R_g)sC_1}{sL_2 + R_g}\right]^{-1} + sL_1$$

$$Z_i = \underbrace{\left\{\frac{sL_2 + R_g}{s^2C_1L_2 + sR_gC_1 + 1}\right\}}_{Z_2} + \underbrace{sL_1}_{Z_{L1}} = \frac{sL_2 + R_g + sL_1(s^2C_1L_2 + sR_gC_1 + 1)}{(s^2C_1L_2 + sR_gC_1 + 1)}$$

$$Z_i = \frac{s^3C_1L_2L_1 + s^2R_gC_1L_1 + s(L_2 + L_1) + R_g}{s^2C_1L_2 + sR_gC_1 + 1}$$

$$Z_i - \underbrace{Z_{01}}_{R_g} = \frac{s^3C_1L_2L_1 + s^2R_gC_1L_1 + s(L_2 + L_1) + R_g - R_g(s^2C_1L_2 + sR_gC_1 + 1)}{s^2C_1L_2 + sR_gC_1 + 1}$$

$$Z_i - Z_{01} = \frac{s^3C_1L_2L_1 + s^2\overbrace{(R_gC_1L_1 - R_gC_1L_2)}^{0,\ L_1 = L_2} + s(L_2 + L_1 - R_g^2C_1) + \cancel{R_g - R_g}}{s^2C_1L_2 + sR_gC_1 + 1}$$

$$Z_i - Z_{01} = \frac{s^3C_1L_2L_1 + s(L_2 + L_1 - R_g^2C_1)}{s^2C_1L_2 + sR_gC_1 + 1} = \frac{s^3 \cdot 2 + s \cdot 0}{s^2 \cdot 2 + s2 + 1} = \boxed{\frac{2s^3}{2s^2 + 2s + 1}}$$

# Ejercicio14_Hoja_2

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$$\cdot\ Z_i + Z_{o1} = \frac{s^3 C_1 L_2 L_1 + s^2 R_g C_1 L_1 + s(L_2 + L_1) + R_g + R_g(s^2 C_1 L_2 + s R_g C_1 + 1)}{s^2 C_1 L_2 + s R_g C_1 + 1}$$

$$Z_i + Z_{o1} = \frac{s^3 C_1 L_2 L_1 + s^2 (R_g C_1 L_1 + R_g C_1 L_2) + s(L_2 + L_1 + R_g^2 C_1) + R_g + R_g}{s^2 C_1 L_2 + s R_g C_1 + 1}$$

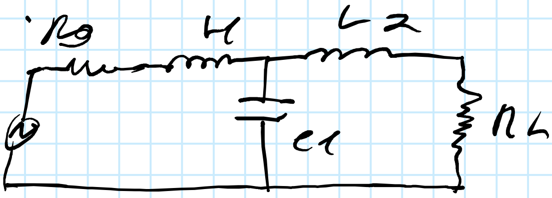
$$\boxed{Z_i + Z_{o1} = \frac{2s^3 + 4s^2 + 4s + 2}{2s^2 + 2s + 1}}$$

$$S_{11} = \frac{\dfrac{2s^3}{\cancel{2s^2+2s+1}}}{\dfrac{2s^3+4s^2+4s+2}{\cancel{2s^2+2s+1}}} = \boxed{\frac{2s^3}{2s^3+4s^2+4s+2} = S_{11}}$$

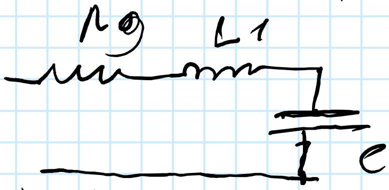
- Por simetría, se demuestra que $S_{11} = S_{22}$

- $S_{21} = \dfrac{V_2}{V_g} \cdot 2\sqrt{\dfrac{Z_{o1}}{Z_{o2}}}$ , $Z_{o1} = Z_{o2} = R_g = R_L$

- $S_{21} = \dfrac{V_2}{V_g} \cdot 2$ , Busco [T] de



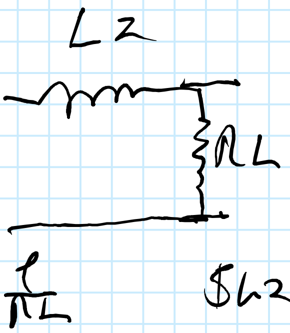

la celda en "L" =




$$\Rightarrow [T_1] = \begin{bmatrix} 1 + Z_1 Y_1 & Z_1 \\ Y_1 & 1 \end{bmatrix}$$

$$[T_1] = \begin{bmatrix} 1 + (R_g + sL_1)\cdot sC_1 & R_g + sL_1 \\ sC_1 & 1 \end{bmatrix}$$

2da celda en "L"




$$\Rightarrow [T_2] = \begin{bmatrix} 1 + Z_2 Y_2 & Z_2 \\ Y_2 & 1 \end{bmatrix}$$

$$[T_2] = \begin{bmatrix} 1 + sL_2 \cdot \frac{1}{R_L} & sL_2 \\ \frac{1}{R_L} & 1 \end{bmatrix}$$

# Ejercicio14_Hoja_3

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$$[T] = [T_1]\cdot[T_2]$$

$$[T] = \begin{bmatrix} 1+(R_g+sL_1)sC_1 & R_g+sL_1 \\ sC_1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1+sL_2\cdot\frac{1}{R_L} & sL_2 \\ \frac{1}{R_L} & 1 \end{bmatrix}$$

$$[T] = \begin{bmatrix} (1+(R_g+sL_1)sC_1)\cdot(1+sL_2\cdot\frac{1}{R_L}) + (R_g+sL_1)\cdot\frac{1}{R_L} & (1+(R_g+sL_1)sC_1)sL_2+1(R_g+sL_1) \\ sC_1(1+sL_2\frac{1}{R_L})+1\cdot\frac{1}{R_L} & sC_1\cdot sL_2+1\cdot 1 \end{bmatrix}$$

$$A = (1+(R_g+sL_1)sC_1)\cdot(1+sL_2\frac{1}{R_L}) + (R_g+sL_1)\cdot\frac{1}{R_L}$$

$$A = (1+sC_1R_g+s^2L_1C_1)\cdot(1+s\frac{L_2}{R_L}) + \frac{R_g}{R_L} + s\frac{L_1}{R_L}$$

$$A = 1+s\frac{L_2}{R_L}+sC_1R_g+s^2C_1R_g\frac{L_2}{R_L}+s^2L_1C_1\cdot 1+s^3L_1C_1\frac{L_2}{R_L}+R_g+s\frac{L_1}{R_L}$$

$$A = s^3L_1C_1\frac{L_2}{R_L}+s^2\left(C_1R_g\frac{L_2}{R_L}+L_1C_1\right)+s\left(\frac{L_2}{R_L}+C_1R_g+\frac{L_1}{R_L}\right)+1+R_g$$

$$A = 2s^3+4s^2+4s+2 = \boxed{(s^3+2s^2+2s+1)\cdot 2 = A}$$

$$\boxed{\frac{1}{A} = \frac{V_2}{V_g}} \Rightarrow$$

$$\cdot\, S_{21} = \frac{V_2}{V_g}\cdot 2 = \frac{1}{(s^3+2s^2+2s+1)2}\cdot 2 = \boxed{\frac{1}{s^3+2s^2+2s+1} = S_{21}}$$

Por ser circuito pasivo, se demuestra que $S_{21} = S_{12}$

# Ejercicio14_Hoja_4

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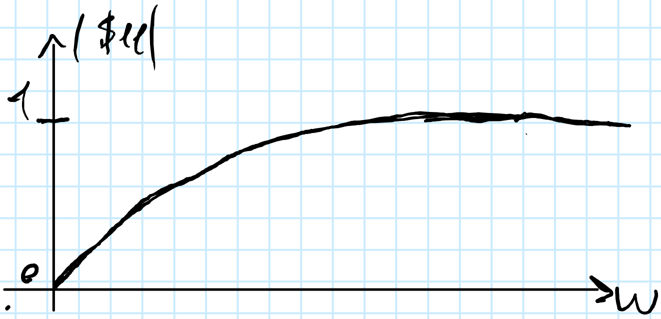
$$\boxed{\frac{2s^3}{2s^3+4s^2+4s+2} = S_{11}} \quad , \quad \boxed{\frac{1}{s^3+2s^2+2s+1} = S_{21}}$$

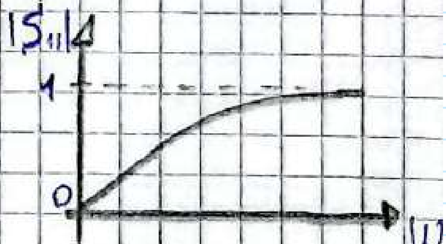




- Del circuito, se observa una red T pasa bajos.
- En la banda pasante $|S_{11}|$ vale cero para $\omega \to 0$ y $|S_{21}| \simeq 1$ para $\omega \to 0$
- Para bajas frecuencias, hay match con respecto a, todo lo que se envía se transmite en términos de potencia

Si $Z_x = R$: $S_{11} = \frac{R}{R + 2 \cdot R_0} = 0 \to$ si $R = 0$

Si $Z_x = SL$: $S_{11} = \frac{SL}{SL + 2R_0}$




Si $Z_x = \frac{1}{SC}$: $S_{11} = \frac{\frac{1}{SC}}{\frac{1}{SC} + 2R_0}$